

R. VIKRAM ADITYA

AI & Backend Engineer

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AI and backend engineer with production experience building RAG pipelines, fine-tuned LLMs, agentic tool-calling, API gateways with rate limiting and failover, and full observability with OTel/Prometheus/Grafana. Proficient in Python, Spring Boot, PostgreSQL, and Azure, with hands-on experience in semantic search, authentication, and data-driven application workflows. Comfortable working across AI experimentation, backend architecture, cloud deployment, and product integration.

EXPERIENCE

Thoughtfocus

Jan 2025 – Present

AI Engineer

- Built an internal HR knowledge assistant over 200+ policy documents using Python, FAISS, Sentence Transformers, and a Qwen 3 model fine-tuned via Unsloth/QLoRA, with semantic chunking, role-based access control, and agentic tool-calling to raise IT tickets and send automated HR notification emails.
- Developed PitStop Manager, an enterprise HR and recruitment platform using React, Spring Boot, Azure Container Apps, PostgreSQL, and Blob Storage, with Azure AD SSO and semantic resume-to-JD matching for automated candidate screening.
- Built Selenium-based automation for Adobe Experience Manager (AEM) forms using Faker-generated dynamic datasets, reducing manual regression testing effort by 60%.
- Mentored an intern on retrieval pipeline development, improving delivery quality and implementation speed.

Tech Mahindra

Nov 2024 – Jan 2025

AI/ML Engineer Intern

- Contributed to Project Indus, Tech Mahindra's enterprise LLM initiative; built and evaluated RAG workflows, benchmarked 5+ model configurations, and improved retrieval relevance by ~30% using Hugging Face Transformers, PEFT, TRL, and bitsandbytes on GPU-backed GCP.
- Built and integrated Django + MySQL backend APIs for an internship application platform, enabling structured data flow between frontend and server-side systems.

Mavencart

June 2023 – Aug 2023

Backend Developer Intern

- Contributed to backend development for two AI products a generative AI writer and a social media template generator, supporting scalable content generation workflows that reduced manual content creation effort by ~40%.

PROJECTS

LLM Sentinel [GitHub] | Python, FastAPI, Redis, OpenTelemetry, Prometheus, Grafana

- Built an LLM API gateway with Redis-backed rate limiting/budgets, circuit-breaker failover, and OTel/Prometheus/Grafana observability runs at \$0 via Ollama and real-wire-format provider mocks.

Intelligent Tutoring System [GitHub] | Python, FastAPI, React

- Full-stack learning platform that scores quizzes, recommends tiered learning resources, and summarizes PDFs via PyMuPDF/NLTK.

Resume Summarizer [GitHub] | Python, Django, React

- Full-stack resume analysis tool that extracts resume sections via GPT and generates summaries and interview questions.

EDUCATION

MVJ College of Engineering

Bangalore, Karnataka

Bachelor of Engineering in Artificial Intelligence and Machine Learning

2020 – 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Ruby, SQL, C++

AI & Machine Learning: RAG, LLM Fine-Tuning, Hugging Face Transformers, FAISS, Sentence Transformers, PEFT, TRL, Unsloth, QLoRA, Ollama, TensorFlow, Keras, scikit-learn

Backend & Web: Django, Spring Boot, FastAPI, Flask, Ruby on Rails, Node.js, REST APIs, React, Streamlit

Cloud & Databases: Azure (Container Apps, Blob Storage, Active Directory, Application Insights, PostgreSQL), MySQL, Google Cloud Platform

Libraries & Automation: NumPy, pandas, OpenCV, Matplotlib, NLTK, Selenium, Playwright, Faker, Git